

Correcting Methylation Calls in Clinically Relevant Low-Mappability Regions

Caiden M. Kumar¹, Devon P. Ryan², Bradley W. Langhorst¹

November 6, 2025

1. New England Biolabs
2. GeneData AG

Abstract

DNA methylation is an important regulatory component in vital biological functions such as embryonic development, and carcinogenesis. Accurate methods to assess genomic methylation status are crucial to its effective use in many scenarios, especially in the detection and diagnosis of disease. Methylation-aware aligners, such as Bismark and bwa-meth, frequently assign significantly higher MapQ values than can be supported by the uniqueness of the region reads are mapped to. These incorrectly high MapQs result in inappropriate methylation calling in repetitive regions. We observe reads that should map to separate locations (possibly having different methylation states) actually end up mapping to the same locus, causing apparent mixed methylation at such loci. Methylation calling can be improved by using Bismark mappability data to filter out insufficiently unique reads. However, simply filtering out Cs in insufficiently unique regions is not adequate as it unnecessarily filters Cs in small mappability dips. These Cs can in fact often be called using reads anchored in a nearby mappable region. We have created a new feature for the MethylDackel methylation caller to perform read-based filtering. This new methylation calling method removes many unsupported methylation calls across the genome. We examined data from a clinical study and identified 228-301 thousand such CpG calls (44-48 thousand in clinically relevant genes) in their dataset using the T2T v2.0 reference genome and the bwa-meth aligner. Left uncorrected, these unsupported calls could result in false identification of methylation markers associated with disease states that in fact do not exist. The presence of such erroneous data in training sets for machine learning models has been shown to reduce performance and can be avoided through the use of this updated calling method.

Introduction

As DNA methylation status can have a significant biological function [33], it is important that there be an accurate way of calling methylation on a genome. Although there are multiple varieties of DNA methylation, perhaps the most significant type in eukaryotes is methylation of cytosine to 5-methylcytosine [2]. Data on DNA cytosine methylation state can be gathered using a methylation sequencing technique (Figure 2), like bisulfite sequencing [4]. In bisulfite sequencing, unmethylated cytosines are deaminated to uracil by the addition of sodium bisulfite. 5-methylcytosines are not affected. Since uracil sequences as thymine and 5-methylcytosine sequences as cytosine, positions of unmethylated Cs in a reference sequence can be identified by C->T transitions[4].

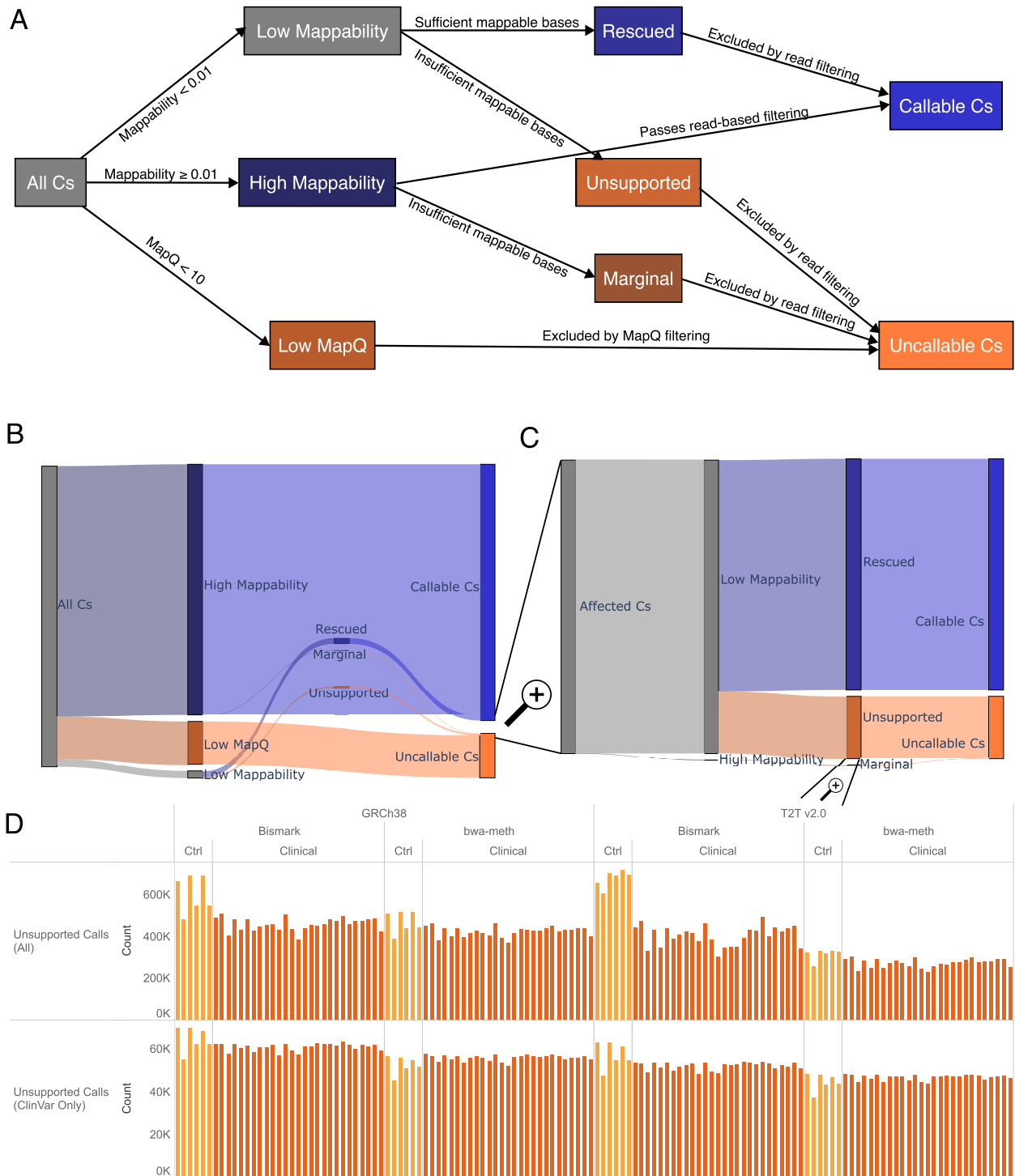


Figure 1: Overview of filtering Cs with our read-based filtering technique. Section A shows the paths Cs can take through our analysis pipeline, leading to either being called or discarded. Section B shows the relative fractions of Cs passing through each step. Section C shows the same data as section B, but with the overwhelming numbers of callable Cs in high-mappability regions and Cs only called from low-MapQ reads excluded to focus on the outcomes more relevant to our method. Section D shows the number of unsupported calls split by aligner and reference, with one bar for each cell line control or clinical library. Without our filtering method all libraries we analyzed were polluted by thousands of unsupported calls.

It is also possible to use an enzymatic method, EM-seq, employing TET2 to oxidize 5-methylcytosine and an APOBEC enzyme to deaminate unmodified cytosine to uracil. While sodium bisulfite treatment produces double strand breaks and other DNA damage, this enzymatic method deaminates with more precision [36], producing longer, better mappable fragments.

Whichever method is used to deaminate cytosines, methylation sequence data is typically aligned to a reference genome using a methylation-aware aligner [9], which is specifically designed to handle the C->T transitions in methylation sequencing data when aligning the reads to a reference. Once aligned, a methylation caller is needed to analyze alignments and report the methylation status of particular cytosines (Figure 2). The resulting data shows the methylation status of each callable cytosine in the genome and can therefore be used to find and study biologically significant DNA methylation sites.

There are two commonly used methylation callers, MethylDackel [32] and bismark_methylation_extractor [18]. The Bismark aligner creates XM, XR, and XG tags that are used by the bismark_methylation_extractor to call methylation. Bismark_methylation_extractor requires these tagged bam files and is not compatible with bam files produced by other aligners. MethylDackel extract is not tightly coupled to a specific aligner or tag scheme. Instead it performs the filtering, quality assessment, and reference comparisons needed for methylation calling internally. MethylDackel efficiently loads a complete or targeted reference genome, examining alignments in chunks that are processed in parallel with all the context needed to differentiate methylation from C->T mutation. Flexible and optional thresholds are available for base-quality, positional trimming, adjacent base context (e.g. CpG or CHH), alignment flags, and MapQ.

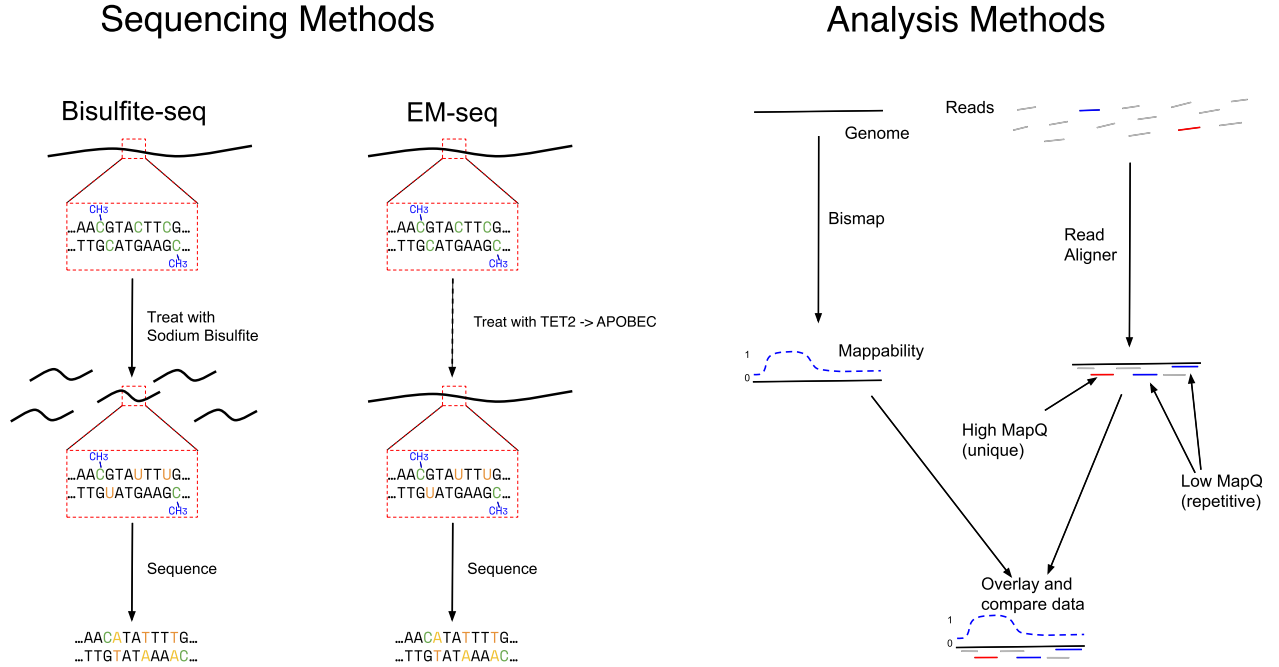


Figure 2: Overview of Methylation Sequencing and Data Analysis Methods.

A read must be unambiguously placed if it is to provide information about a specific region of a genome. Reads that equally match more than one area of the reference genome should not be used to assess methylation of any given C. To avoid calling Cs using reads derived from multiple genomic loci, methylation aligners (and

read aligners in general) assign a MapQ value to each read alignment (Figure 2). According to the SAM specification [11], MapQ is defined as: “ $-10 \log_{10} \Pr\{\text{mapping position is wrong}\}$, rounded to the nearest integer”. MapQ indicates how uniquely placed a read alignment is, that is, in how many other places could the read align to the reference. A low MapQ means that the read may align in many places throughout the genome (for instance, a read of centromeric satellite DNA would likely have a very low MapQ). A high MapQ indicates that the read likely aligns where it is placed and nowhere else in the genome.

A methylation caller can use accurate MapQ values to filter out reads with multiple placements in the genome, allowing the resulting methylation calls to accurately reflect their specific loci. Unfortunately, current aligners frequently produce implausible MapQ scores, and as a result methylation callers can produce unsupported methylation calls. Such data could result in the identification of invalid biomarkers, and if used as training data for a machine learning model, can reduce its performance[8].

In this paper, we describe the problem that low-mappability methylation calls create for methylation calling, the extent of wrongly called methylation in real datasets, and our implementation of an improved method that measurably reduces unsupported methylation calls improving differential methylation assessment, the use of methylation as a biomarker for disease progression, and understanding of development processes.

Results

While evaluating the methylation aligner bwa-meth [27], we observed a significant number of reads with unexpectedly high MapQ values in repetitive regions (e.g. centromeres, Figure 3). Another commonly used aligner Bismark[19] produced even more calls in repetitive regions (data not shown). After observing high MapQ reads in the centromere and larger repetitive regions, we investigated to see if smaller regions might also be too repetitive to support the high aligner MapQ estimates we observed. We identified repetitive regions using data from Bismap [15], a tool that counts the number of occurrences of every single K-mer of a particular length (in this case, $k=100$) in the genome to create a mappability score (ranging from 0 to 1 in increments of 0.01) for every base in the reference. Each individual K-mer is considered either mappable or not, and the overall mappability score is derived (indirectly) from how many different K-mers are mappable and how many are not. Bismap takes the effect of C->T conversion into account and therefore produces data which is applicable in the context of methylation sequencing [15]. Reads entirely contained within a region of low mappability should not have high MapQ values due to their repetitiveness, however we observed many high-MapQ reads in regions with very low or zero Bismap mappability (e.g. Figure 4).

While low MapQ can indicate repetitiveness, even stringent MapQ thresholds cannot reliably select reads for safe methylation calling in regions containing repetitive DNA (Figure 5). We considered excluding methylation calls on Cs found in low-mappability regions (e.g. using bedtools), but rejected this approach because it is prone to discarding Cs which could potentially be rescued. In addition, Cs in short, unique regions or on the boundary of larger mappable regions (marginal Cs) would be kept even if the surrounding DNA is repetitive (Figure 6), but this situation was not commonly observed. Discarding salvageable Cs in short repetitive regions is a larger problem. In this scenario, a C located in a small dip in mappability would be eliminated despite coverage from read pairs anchored in nearby unique regions (Figure 7). When calling using EM-seq reads aligned with bwa-meth to T2T, marginal Cs are rare (in CpG contexts, between 0.000198% and 0.000245% of genome CpGs, or 134-166 CpGs) but discarding salvageable Cs is much more common (in CpG contexts, between 1.57% and 1.81% of genome CpGs, or 1.07-1.23 million CpGs,) (Figure 5). In cases where reads from multiple low-mappability regions are placed onto one region (creating a coverage spike), filtering by coverage could possibly also remove the problematic region, but in cases where reads are simply mixed among multiple low-mappability regions (not creating a coverage spike), coverage filtering would not be effective at removing the affected regions. In addition, if some of the reads used to call a C

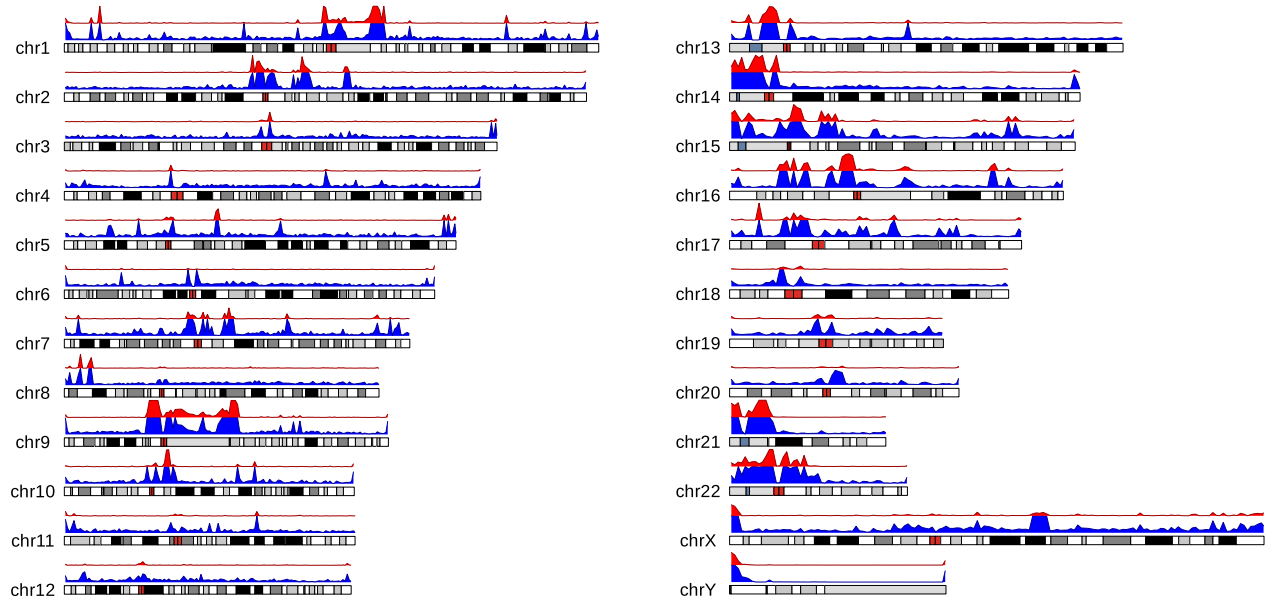


Figure 3: Cs in low mappability regions (blue), as well as those filtered out by read-based filtering that still passed MapQ filtering (red), are represented across all chromosomes. Read-based filtering removes unsupported calls while retaining millions of Cs that a naïve coordinate filtering approach would needlessly eliminate. In this figure, the horizontal banded rectangles represent the chromosomes, the blue track above them shows all calls in low-mappability regions, and the red track corresponds to uncalleable Cs accurately removed by read-based filtering.



Figure 4: While MapQ filtering does not remove all poorly mapped reads, simple coordinate filtering incorrectly discards many salvageable CpGs. Section A shows the number of rescued CpGs across all libraries in our analysis, in the same manner as unsupported calls are shown in Figure 1. Section B shows an example of this issue in a histone H2B gene. The MapQ Filtered track shows CpGs called with only MapQ filtering applied. The Coord. Kept track shows those Cs that would be retained with coordinate filtering. The Rescued track shows mappable CpGs in small mappability dips that coordinate filtering would unnecessarily discard, but our read-based filtering is able to save. Finally, the Read Filtered track shows all the CpGs our method is able to retain - a significantly larger number than with coordinate filtering.

are uniquely mappable, but others are not, read-based filtering can remove only those reads that are not uniquely mappable, rather than being forced to discard the entire locus as is required for coordinate filtering.

To reliably filter out only problematic read pairs (those where both mates are placed in low mappability regions) we modified the MethylDackel methylation caller to accept a bigWig file of low-mappability regions to exclude from analysis. This read-based filtering approach precisely eliminates only those reads in repetitive regions and has a negligible effect on execution speed (see computational methods for details). As this method is able to retain many salvageable Cs in low-mappability regions, they will be hereafter referred to as “rescued Cs”.

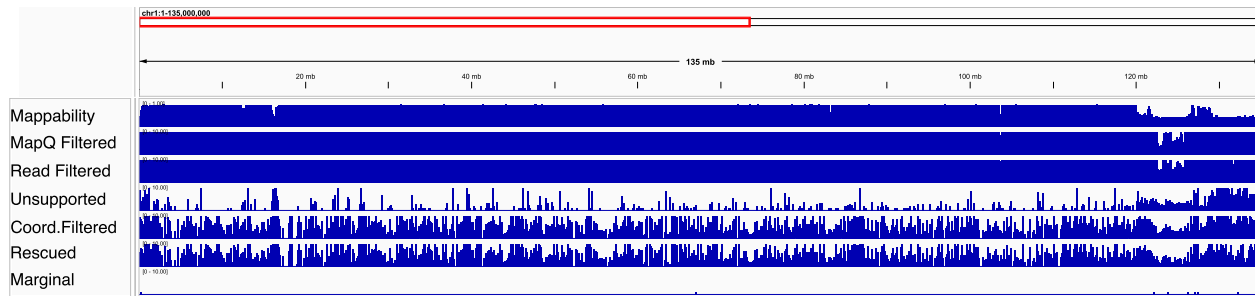


Figure 5: An overview of the impact of the various filtering scenarios in the q arm of T2T v2.0 chromosome 1. The six tracks show the effect of coordinate and read-based mappability filtering on the methylation calls on this chromosome arm and are representative of rest of the genome. “Mappability” is the Bismap mappability for the chromosome. “MapQ Filtered” is the calls after commonly used MapQ filtering only (no coordinate or read-based mappability filtering). “Read Filtered” shows the methylation calls kept after read-based mappability filtering. “Unsupported” shows the calls in low-mappability regions removed entirely by read-based filtering for being insufficiently mappable. “Coord. Filtered” shows the methylation calls kept after coordinate-based (i.e. bedtools) filtering. “Rescued Cs” are sufficiently mappable according to read-based filtering. Most of the Cs removed by coordinate filtering are rescued, producing similar tracks. In the last track, a small number of “Marginal Cs” are not sufficiently mappable according to read-based filtering, but are retained by coordinate filtering.

To focus on the incorrect alignments and methylation calls that have biological and medical significance, we examined methylation calls in and 2 kb upstream of GENCODE genes [12] that contain variants listed in the ClinVar [20] database of disease-associated variants. Such genes will be hereafter referred to as “ClinVar genes”. Using our alignment filtering approach we successfully avoided wrongly calling 46,570 CpGs in these important regions which have insufficient mappability to support methylation calling in our highest quality dataset (a 200 ng EM-seq sample aligned to the T2T reference with bwa-meth). Briefly, to identify the number of methylation calls correctly removed, we counted unfiltered CpGs in our regions of interest, subtracting that from all CpGs we kept after read-based filtering. The result was a count of CpGs that were correctly filtered out (Figure 1). This count is a combination of unsupported CpGs (CpGs removed by read-based filtering in low-mappability regions) and marginal CpGs (CpGs removed by read-based filtering in high-mappability regions), and will be hereafter referred to as “problematic CpGs”.

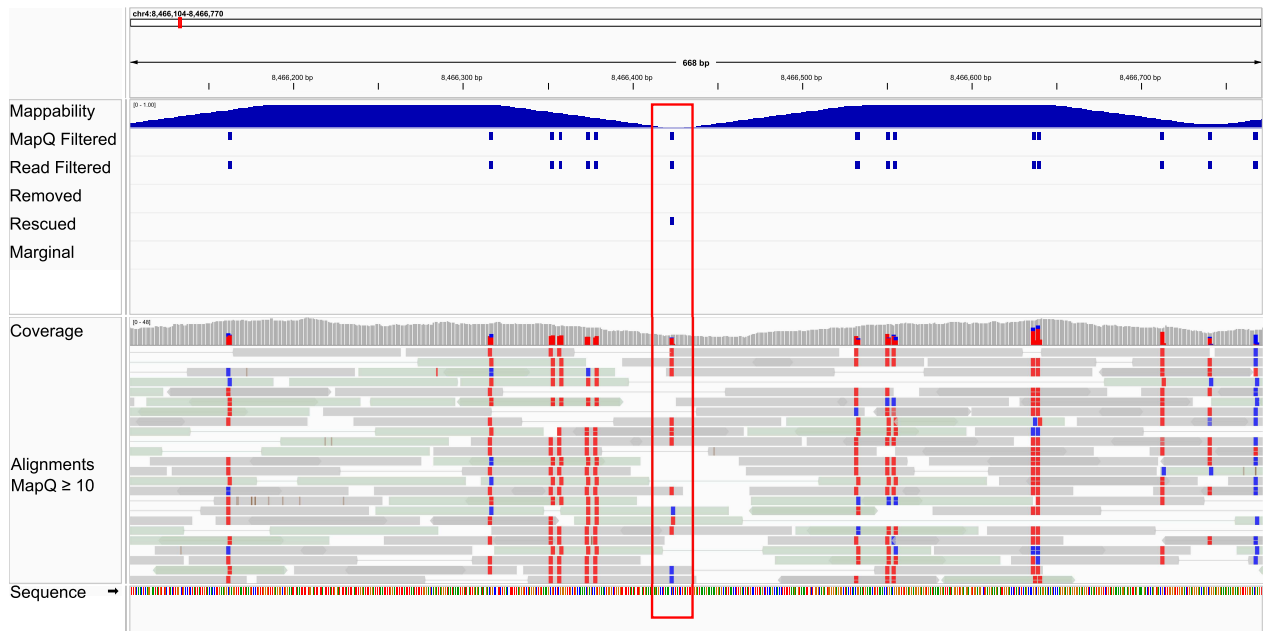


Figure 6: An example of a C incorrectly discarded by coordinate filtering (red box). The C is in a small mappability dip, causing it to be discarded by bedtools filtering, while read-based filtering correctly retains the C, as it can be called using reads extending into the high-mappability regions on either side.

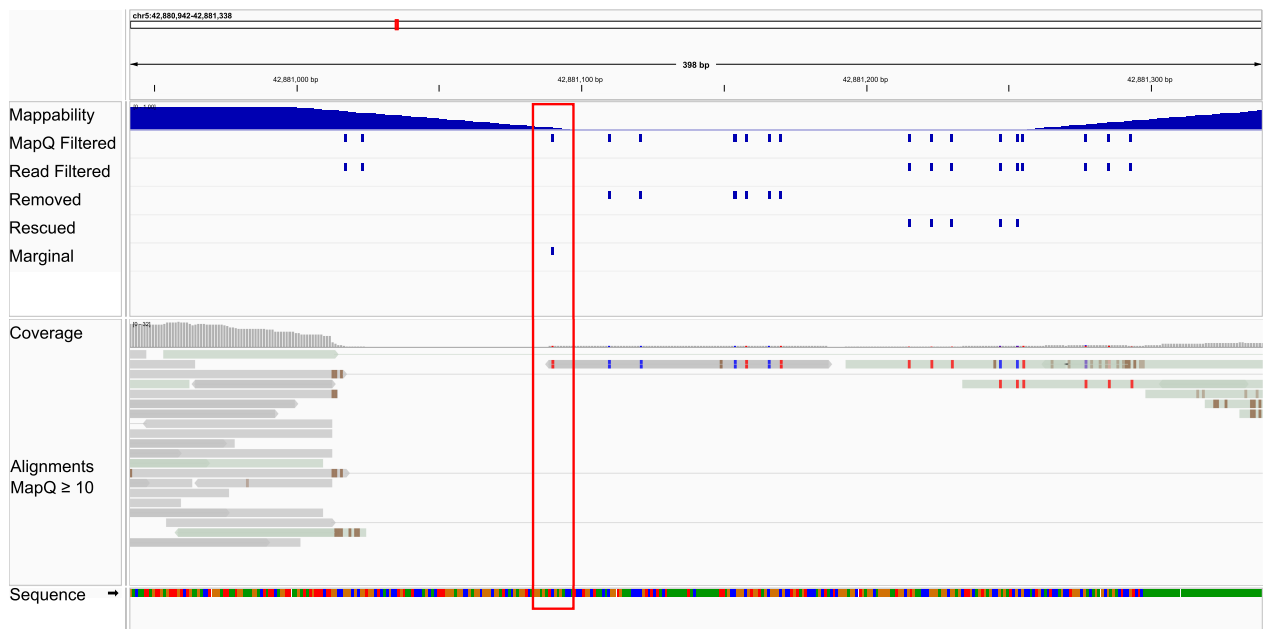


Figure 7: An example of a C incorrectly kept by coordinate filtering (red box). As the C is itself just inside the edge of a mappable region, bedtools retains it, even though it is only supported by two insufficiently mappable reads. Our read-based filtering correctly removes this sort of C, as there is not enough data here to call it confidently.

Filtering Effect

We compared the effect of read-based mappability filtering on number of CpG's called, as well as number of CpG's called in low-mappability regions (as defined by coordinate filtering), on EM-seq libraries of 3 sample masses aligned with the bwa-meth aligner. We saw a significant positive filtering effect on all 3 sample input amounts, with a mean of 328,199 CpGs filtered out by read-based filtering in each sample (see Figure 1). These problematic Cs were scattered across the genome, as shown in Figure 3.

Effect on Medically Relevant Genes

To estimate biological significance of unsupported CpGs, we used genes in GENCODE containing variants in ClinVar ("ClinVar genes"). Focusing on genes that may contain unsupported calls, we filtered the list to only include those genes which had at least one low-mappability region, either in the gene or in a 2 kb region upstream of the annotated start site (to capture regulatory regions). Examination of this data numerically and in IGV[31] revealed that there were unsupported calls associated with thousands of clinically relevant genes in the 200 ng EM-seq/bwa-meth data, with each gene containing 1-4,276 unsupported CpG calls, with a median of 5 unsupported CpGs per gene. This indicates that indeed unsupported calls may affect biologically significant methylation. An example of unsupported calls from a 200 ng EM-seq sample aligned to T2T using bwa-meth are provided in Figure 8.



Figure 8: A view of a cluster of unsupported calls near the boundary between high and low mappability regions in the gene PDE4DIP (ENSG00000178104.19), regulation of which is reported to relate to dementia[28], schizophrenia[3], pineoblastoma[34] and other human disease[26, 1]. The read alignments (MapQ ≥ 10) are shown paired, with a line connecting reads 1 and 2, and highlighted for CpG methylation. Note that some methylation calls in the low-mappability region are not unsupported calls, as they can be called using read pairs extending into a high-mappability region.

Clinical Samples

To further demonstrate the clinical impact of improperly calling methylation in repetitive regions, we also examined sequence data from 27 clinical samples from [6]. This study sought to examine genetic markers connected to the risk of developing type 2 diabetes and insulin resistance. DNA methylation data was collected by the study authors using whole genome bisulfite (WGBS) sequencing on blood samples from the enrolled patients, sequenced on an Illumina HiSeq X. We ran these datasets in FASTQ format through the exact same analyses as our cell line data, resulting in a total of 108 (27 libraries * 2 aligners * 2 references) processed datasets. Examining the datasets aligned to T2T with bwa-meth, we observed a mean of 273,682 unsupported CpGs across the genome, and 46,660 unsupported CpGs in ClinVar genes.

EM-seq vs. WGBS

Because of the reduced DNA damage and longer possible read lengths that are found in non-bisulfite libraries, we compared our EM-seq cell line libraries with WGBS libraries produced using the same 3 sample masses, to see if there is a difference between the two methods with regard to the number of CpGs affected. WGBS libraries have fewer overall CpGs called[36] and therefore fewer unsupported CpGs at all input masses than EM-seq libraries (Figure 1). We also observed that EM-seq libraries had significantly more rescued CpGs, due to longer inserts allowing for more low-mappability CpGs to be called using reads and mates extending into higher-mappability regions. Specifically, in libraries aligned to T2T using bwa-meth, we observed a mean of 735,314 unsupported calls in CpG contexts in WGBS libraries, and 1,156,457 unsupported CpGs in EM-seq libraries (Figure 1).

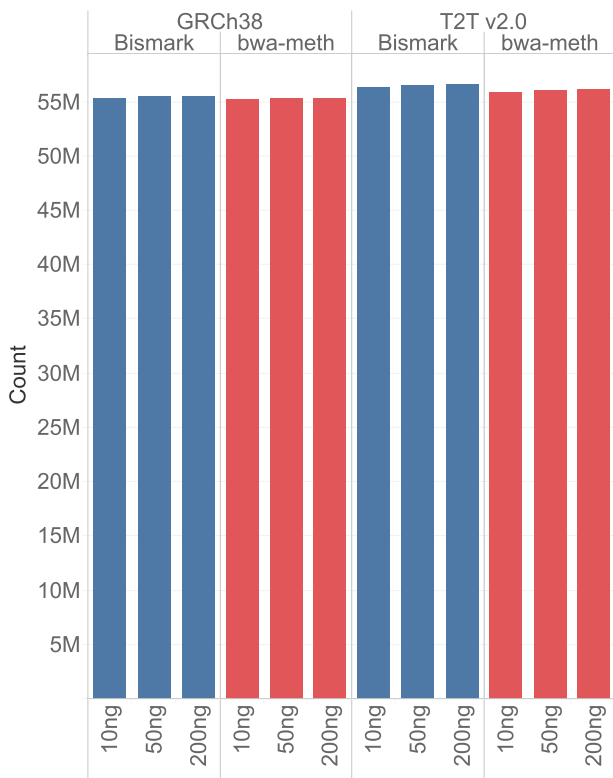
Bwa-meth vs. Bismark

For comparison, we examined the popular Bismark methylation aligner [18] in the same manner as the bwa-meth analysis. We observed that compared to bwa-meth, Bismark produced more inappropriately-high MapQ reads in repetitive regions (Figure 3). In addition, Bismark and bwa-meth use different systems to define values for MapQ. Bismark (as a result of using Bowtie2 for alignment) reports MapQ based on number of reference mismatches, number of alternate alignments, and base qualities, producing values between 0 and 42 [29, 14]. Bwa-meth (using BWA-MEM for alignments) conforms to the SAM specification in estimating probabilities in the MapQ field. Overall, we observed a higher number of unsupported calls with Bismark than we did with bwa-meth, as shown in Figure 1. This difference was noticeable both overall and in ClinVar genes, especially in our cell line data.

GRCh38 vs. T2T

We also examined the effect of using the widely-used GRCh38 reference [25] in place of T2T. The T2T reference is much more complete than GRCh38, including repetitive regions such as centromeres, telomeres, and rDNA, which could play a role in where methylation is called and what areas of the genome have unique enough sequence to support calling. We found that, as we expected, there were more unsupported calls in the GRCh38 data as compared to the T2T data. Considering both strands, total calls were roughly similar between the two reference sequences with more Cs available in the T2T reference. (Figure 9, left). Use of read-based filtering has a smaller effect with the more complete reference, indicating that using the T2T reference can allow for better calling of methylation with fewer unsupported calls.

Total Calls GRCh38 vs. T2T v2.0



Unsupported Calls GRCh38 vs. T2T v2.0

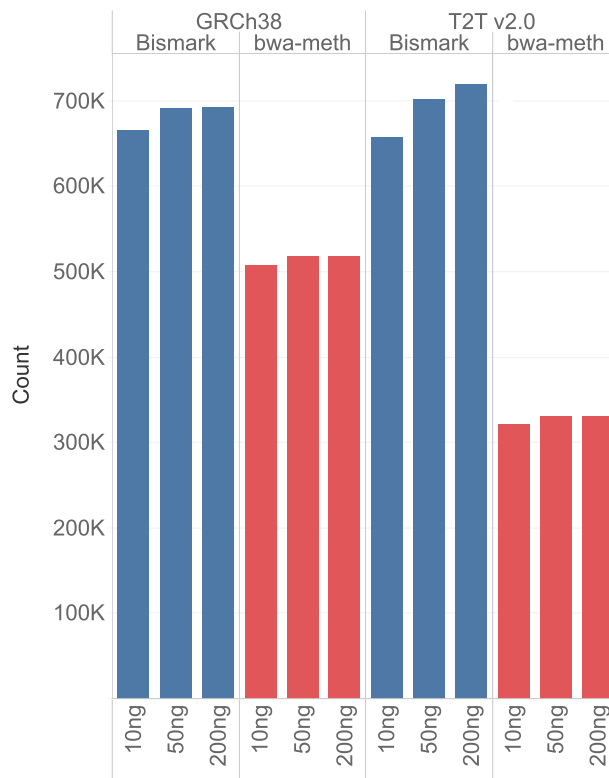


Figure 9: Left: Total CpG counts before filtering in the 10, 50, and 200 ng EM-seq samples, aligned with bwa-meth to GRCh38 and T2T. The vertical axis is the number of calls in the sample. Right: Number of Unsupported CpGs in the same samples, where such Cs are filtered out entirely by read-based filtering.

Materials and Methods

DNA Methylation Sequencing

Materials

Previously published[36] libraries from 10, 50, and 200 ng of genomic DNA from the NA12878 cell line (Coriell) were analyzed. This input DNA was supplemented with a small amount of lambda phage DNA (NEB #N3011), and a pUC19 plasmid (NEB #N3041) treated with M.SssI CpG Methyltransferase (NEB #M0226).

Whole Genome Bisulfite Libraries

The whole genome bisulfite (WGBS) libraries were prepared using the Ultra II DNA library prep kit (NEB E7645) before being Bisulfite converted using the Zymo EZ DNA Methylation-Gold bisulfite conversion kit according to the protocol described in Vaisvila et al. [36].

EM-seq Libraries

The EM-seq libraries were prepared using the NEBNext Ultra II DNA library prep kit (NEB E7645) (using the NEBNext EM-seq adapter), then EM-seq converted using TET2 and ABOBEC3A, as described in Vaisvila et al. [36].

Sequencing

Methylation libraries were sequenced with more diverse libraries (~10%) on 2 flow cells of an Illumina NovaSeq 6000 [13] using the S4 chemistry. We analyzed about 1.3 billion 99 bp paired-end reads for the for each library.

Computational Methods

The following data and tools were used for the analysis:

The T2T v2.0 analysis set (hereafter referred to as “T2T”) supplemented with phage T4, phage lambda, phage Xp12, and pUC19 contigs was used throughout [22]. A VCF file of disease-associated variant sites was downloaded from ClinVar (see supplemental data for file date) and a GFF file of the GENCODE gene annotations were used (see supplemental data for file date). For the GRCh38 comparison, the GRCh38 [5] genome was used instead of T2T, supplemented with the same control contigs. The T2T mappability data was a 100bp multi-read bigWig file generated using Bismap on the T2T v2.0 genome reference. The GRCh38 Bismap was the 100bp multi-read bigWig file downloaded from Bismap’s website (see supplemental materials for link).

The Nextflow scripts used to analyze this data can be found at https://github.com/nebiolabs/low_bismap_methyl_calls. A detailed diagram of the analysis pipeline is found in Supplemental figures S.1 and S.2. Tools used include bedtools [30], samtools [21], GNU awk, MethylDackel, bigWigToBedGraph [16], BEDOPS [24], GNU sort, GNU head, bwa-meth, and Bismark. GNU sort and head are required since some of the functionality needed (specifically, the ability to parallelize sorting and the ability to use a negative value with the -n option for head to count lines from the end of the file) is not present in BSD sort and head.

Discussion

Reads placed with falsely high confidence elicit cascading detrimental effects on methylation calling, differential methylation assessment, and assessment of phenotypes associated with methylation status. To remove the spurious, unsupported methylation calls that result from such reads, we suggest that reads with both mates in low mappability regions (as determined by Bismap) should be excluded from methylation calling. Additionally, because of the more accurate MapQ values, decreased run time, and more flexibility to separate methylation calling from alignment, we also recommend the use of bwa-meth for alignment and MethylDackel with $\text{MapQ} \geq 10$ and the new mappability feature enabled for high accuracy methylation calling.

Supplemental Materials

Detailed Computational Methods

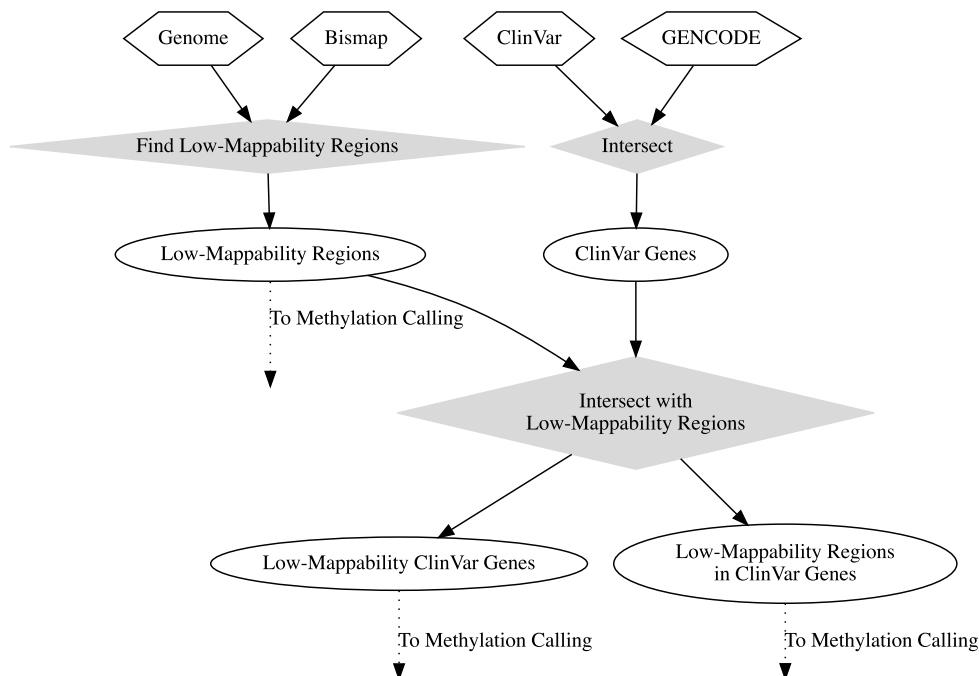


Figure S.1: Detailed description of preprocessing steps used to identify biologically relevant low-mappability regions. Ovals are intermediate files. Gray rhombuses are processing steps with multiple arguments. Dotted lines indicate inputs that go to the methylation calling steps in Figure S.2.

The sequencing reads were aligned to the T2T v2.0 reference genome using the bwa-meth aligner with default options. The input mass (where available), reference, method, and aligner were appended to the filenames for later processing. Methylation calling was performed on the resulting BAMs using MethylDackel

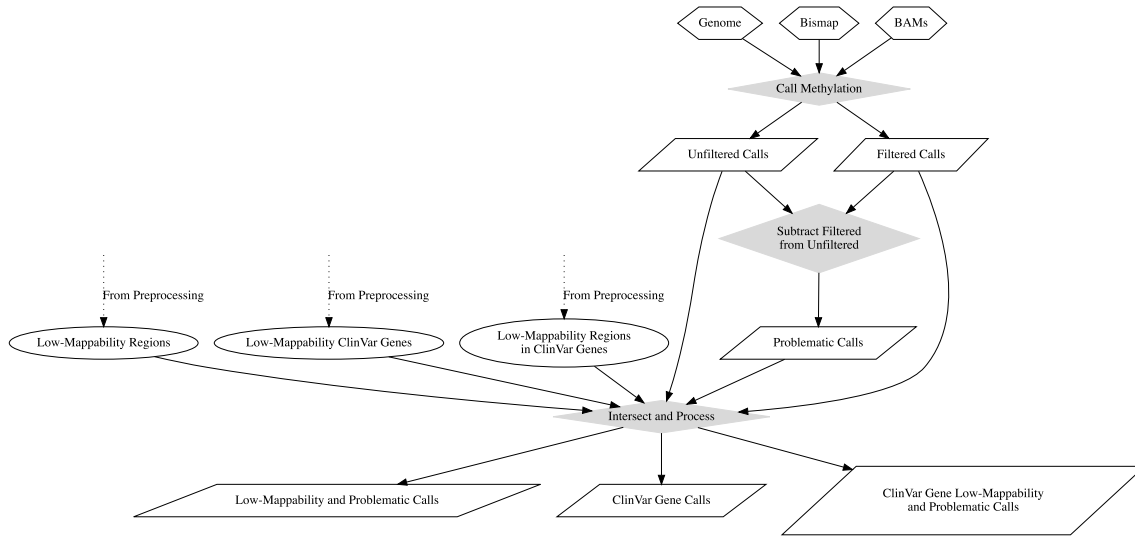


Figure S.2: Detailed description of steps for calling methylation and intersecting the resulting calls with the low mappability regions determined in the preprocessing steps. Ovals are intermediate files. Gray diamonds are processing steps with multiple arguments. Hexagons represent input files. Parallelograms are output files. Dotted lines indicate inputs that come from the preprocessing steps in Figure S.1.

v0.6.1. MethylDackel uses a default value of 10 as a minimum MapQ which should include mostly single locus reads, however inaccurate MapQs can lead to reads being incorrectly included in methylation calling. In order to eliminate reads in low-mappability regions, a feature was created for MethylDackel which allows it to take as an input a bigWig file which is then used to filter out read pairs (this feature currently only supports paired-end reads) where neither mate intersects a high-mappability region. The feature allows for user configuration of the low mappability threshold and the number of bases which must be equal to or above that threshold in order for the read pair to be kept (the defaults are a low mappability threshold of 0.01 and to require 15 bases that are greater than or equal to that threshold in a single read). The filtering algorithm has been optimized both by loading the mappability data into memory before calling, and through the use of a custom run-length compressed binary file format that we here term BBM (Binary BisMap). This format can store the mappability data for the hg38/GRCh38 human genome in 143 MB, compared to 1.11 GB when stored as a bigWig, giving a compression ratio for this dataset of 7.78:1. MethylDackel's trimming options were set to exclude the first 8 bp of read 1 and the last 8 bp of read 2.

The ClinVar VCF and GENCODE GFF were combined using bedtools and BEDOPS into a single BED file listing all GENCODE annotation regions that overlap one or more ClinVar variants (these regions will be referred to as “ClinVar regions”), which will be called the “ClinVar regions BED”. Each region in the ClinVar regions BED was expanded 2 kb upstream using bedtools slop to capture methylation upstream of the gene. The Bismap bedGraph (which did not contain zeroes) and an FAI index of the GRCh38 reference genome FASTA file were processed with bedtools, awk, and GNU sort to obtain a bedGraph containing all mappability data, including zeroes. This file will be referred to as the “Bismap complete bedGraph”. It was then combined with the ClinVar regions BED using bedtools map to create a file containing the minimum and mean mappability for each gene with a ClinVar variant.

The Bismap complete bedGraph was then filtered using awk to produce a file containing only low mappability

bility regions (mappability < 0.01). The file with minimum and mean mappability for every ClinVar region was filtered likewise on minimum mappability. The two resulting files (one of low-mappability regions, one of ClinVar regions with low minimum mappability) were combined to produce a file of all low mappability regions that are in ClinVar regions.

Alignments of paired-end EM-seq and whole genome bisulfite reads were processed using MethylDackel (with and without our enhancement) using a minimum MapQ cutoff of 10 and the default settings mentioned above and combined with the file of all low mappability regions in ClinVar regions to produce a list of all methylation calls in low mappability ClinVar regions (this will be referred to as the “low mappability ClinVar calls file”). Each call was tagged with its methylation context (CpG, CHG, or CHH) for further analysis. A low-mappability methylation call, as used here, is defined as a methylation call that is in a region with Bismark mappability less than 0.01, as defined by the position of the C alone (positions of anchoring reads are not considered when determining this).

The low mappability ClinVar calls file was intersected with the file of ClinVar regions with low minimum mappability to produce a file of ClinVar regions with low mappability calls. The `-wa` option for `bedtools intersect` was used here, which writes a copy of the ClinVar region to the output file for each low mappability call in the region, in order that this file would contain multiple copies of each region, one per low mappability call in the region. These duplicates were then used to count low mappability calls by feeding the data to a custom Python script (found in the Nextflow script in the supplemental materials) which counted and combined the duplicates, producing a list of all ClinVar regions with low mappability calls and how many low mappability calls are in each region. A similar intersection was performed for all ClinVar regions with low minimum mappability (i.e. ClinVar regions that could potentially be affected) to produce a file of all calls (regardless of mappability of the C) in such regions, and a list of such regions with a count of calls per region.

Since this analysis was run for three different input masses and two sequencing protocols, the low mappability calls files were also processed through a custom Python script (see supplemental materials) to add a field specifying which input mass, sequencing protocol, and MethylDackel filtering setting were used. This data was extracted from a suffix appended to the BAM and BED file names. The same field was added to the lists of all ClinVar regions with low mappability calls and counts described previously. We also produced similarly-annotated files for all methylation calls (regardless of mappability), all methylation calls in GENCODE genes (with low minimum mappability) which contained ClinVar variants, all methylation calls in low-mappability regions that were also in GENCODE genes which contained ClinVar variants, counts of calls in in GENCODE genes which contained ClinVar variants regardless of mappability.

To assess problematic calls (i.e. which Cs are filtered out by read-based filtering), as well as under/over-filtering with respect to coordinate filtering (discussed above), pairs of these files were compared using `bedtools` to produce files showing the difference between filtered and unfiltered data. As the raw data files were very large, a Nextflow pipeline was used to produce summary files containing total counts of methylation calls for analysis. All files were examined and compared in Tableau®.

To compare the behavior of Bismark with `bwa-meth`, this analysis was re-run with the Bismark aligner using default settings and deduplicated using `deduplicate_bismark` according to the documented protocol [17].

To compare the need for and the effect of mappability filtering on the more widely-used GRCh38 reference genome [25] (hereafter referred to as “GRCh38”), the analysis was re-run using the GRCh38 reference in place of T2T (supplemented with phage T4, phage lambda, phage Xp12, and pUC19 just as with T2T). In total, this analysis was run on all combinations of sequencing method (WGBS or EM-seq), aligner (Bismark or `bwa-meth`), input mass (10, 50, or 200 ng), filtering (read-based filtering or no read-based filtering), and reference genome (GRCh38 or T2T), using a Nextflow[7] pipeline (see supplemental materials) which managed the

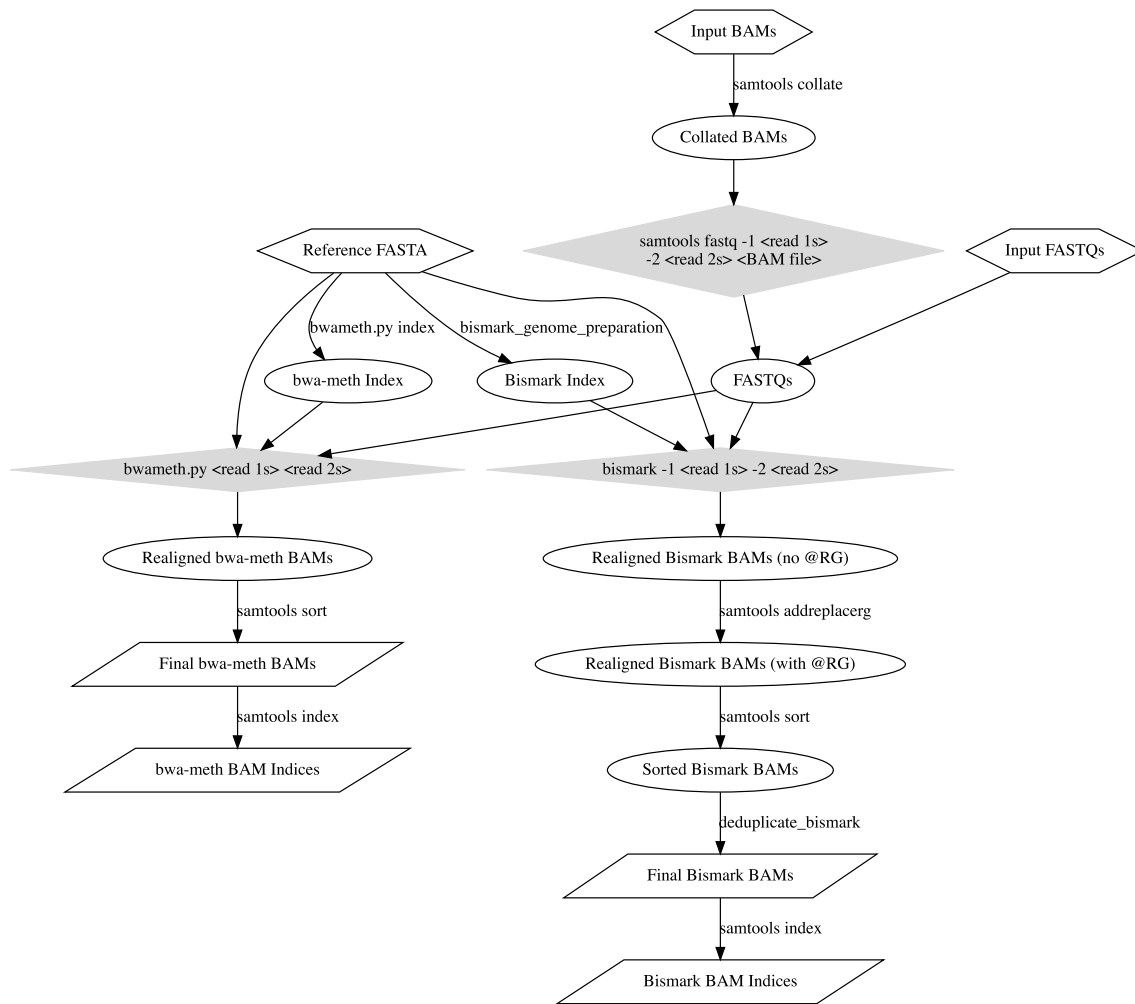


Figure S.3: Detailed description of BAM realignment steps. Hexagons are input files. Ovals are intermediate files. Gray diamonds are processing steps with multiple arguments. Parallelograms are output files. Labels on edges are processing steps with one input and one output.

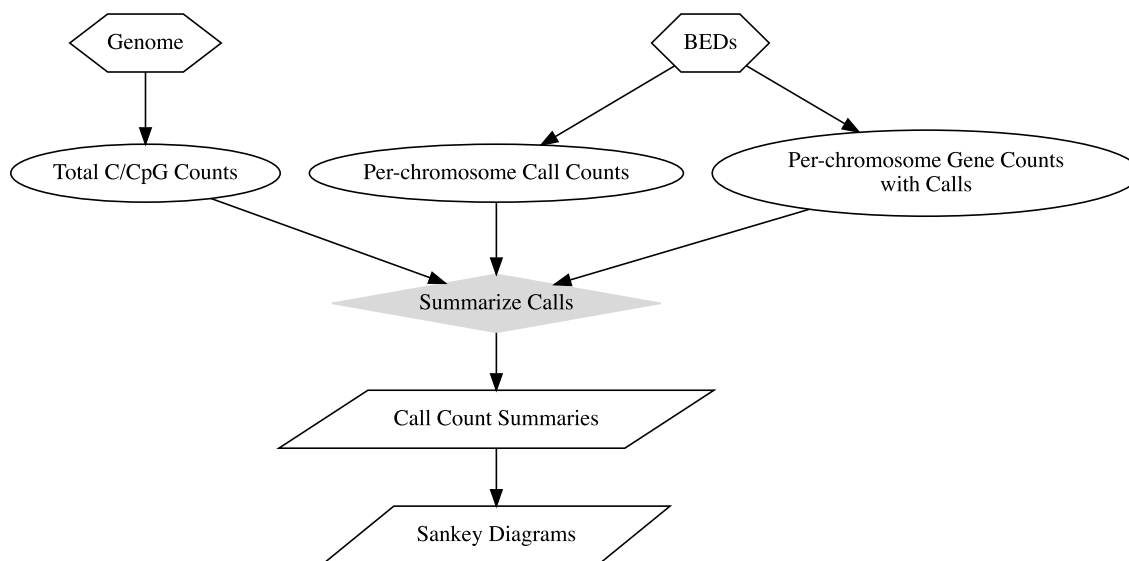


Figure S.4: Detailed description of postprocessing steps. Hexagons are input files. “BEDs” refers to the BED-like tabular files output by the main processing pipeline (Figure S.2). Ovals are intermediate files. Gray diamonds are processing steps with multiple arguments. Parallelograms are output files. Labels on edges are processing steps with one input and one output.

execution of all the analysis tools needed.

External Analysis Resources

Individual tools used in the analysis include samtools, bowtie2, gzip, bgzip, tabix, seqtk, fastp, sambamba[35], bigWigToBedGraph[16], bedGraphToBigWig, MethylDackel, sed, awk, python, GNU sort, GNU head, zstd, bc, plotly, bedtools[30], and BEDOPS[24]. Versions of all tools used are specified in the nextflow DSL2 scripts using conda[23] dependency resolution.

The custom Python script that adds the input mass, sequencing protocol, MethylDackel filtering setting, and aligner name can be found at https://github.com/nebiolabs/low_bismap_methyl_calls.

The pull request for the patch adding mappability support to MethylDackel can be found at <https://github.com/dpryan79/MethylDackel/pull/80>. It was merged into MethylDackel in version 0.5.0.

The BBM format definition is available at https://github.com/dpryan79/MethylDackel/blob/master/BBM_Specification.md.

The Bismap file used for this analysis was downloaded from <https://www.pmgenomics.ca/hoffmanlab/proj/bismap/trackhub/hg38/k100.Bismap.MultiTrackMappability.bw>

The GRCh38 ClinVar VCF was the March 13, 2022 version of ClinVar’s variants VCF.. The T2T ClinVar VCF was the version lifted over from ClinVar 20220313 by the T2T Consortium and available on their GitHub page [22].

The GRCh38 GENCODE GFF was the GENCODE v35 “comprehensive gene annotation” CHR GFF3 file. [22] The T2T GENCODE GFF is the Liftoff-converted version of GENCODE v35 from the T2T Consortium’s GitHub [22].

The chromosome ideogram figure was generated using karyoploteR[10]. The T2T cytobands used were obtained from the T2T Consortium’s GitHub page [22].

Acknowledgments

We are grateful to the many people who contributed to the EM-seq research and product development project and made these libraries available to the public. This work would not have been possible without access to an extensible, open-source methylation caller. Many thanks to Chaithanya Ponnaluri for critical reading of this manuscript, to Matt Campbell and Ariel Erijman for helpful discussion, and to the organizers and supporters of the summer internship and postbac program at New England Biolabs that enabled this project.

References

- [1] Paul L. Auer et al. “Rare and Coding Region Genetic Variants Associated With Risk of Ischemic Stroke: The NHLBI Exome Sequence Project”. eng. In: *JAMA neurology* 72.7 (July 2015), pp. 781–788. ISSN: 2168-6157. DOI: 10.1001/jamaneuro.2015.0582.
- [2] Achim Breiling and Frank Lyko. “Epigenetic regulatory functions of DNA modifications: 5-methylcytosine and beyond”. In: *Epigenetics & Chromatin* 8 (2015), p. 24. ISSN: 1756-8935.
- [3] L. M. Camargo et al. “Disrupted in Schizophrenia 1 Interactome: evidence for the close connectivity of risk genes and a potential synaptic basis for schizophrenia”. eng. In: *Molecular Psychiatry* 12.1 (Jan. 2007), pp. 74–86. ISSN: 1359-4184. DOI: 10.1038/sj.mp.4001880.
- [4] Gary G. Chen et al. “Medium throughput bisulfite sequencing for accurate detection of 5-methylcytosine and 5-hydroxymethylcytosine”. In: *BMC Genomics* 18 (Jan. 2017). 28100169[pmid], p. 96. ISSN: 1471-2164.
- [5] Deanna M. Church et al. “Modernizing reference genome assemblies”. eng. In: *PLoS biology* 9.7 (July 2011), e1001091. ISSN: 1545-7885. DOI: 10.1371/journal.pbio.1001091.
- [6] *dbGaP Study*. URL: https://www.ncbi.nlm.nih.gov/projects/gap/cgi-bin/study.cgi?study_id=phs001048.v2.p1 (visited on 09/27/2024).
- [7] Paolo Di Tommaso et al. “Nextflow enables reproducible computational workflows”. en. In: *Nature Biotechnology* 35.4 (Apr. 2017), pp. 316–319. ISSN: 1546-1696. DOI: 10.1038/nbt.3820. URL: <https://www.nature.com/articles/nbt.3820> (visited on 07/12/2021).
- [8] Farzaneh S. Fard et al. “Impact of biased mislabeling on learning with deep networks”. en. In: *2017 International Joint Conference on Neural Networks (IJCNN)*. Anchorage, AK, USA: IEEE, May 2017, pp. 2652–2657. ISBN: 978-1-5090-6182-2. DOI: 10.1109/IJCNN.2017.7966180. URL: <http://ieeexplore.ieee.org/document/7966180/> (visited on 09/27/2024).
- [9] Francine E. Garrett-Bakelman et al. “Enhanced Reduced Representation Bisulfite Sequencing for Assessment of DNA Methylation at Base Pair Resolution”. In: *J Vis Exp* 96 (Feb. 2015). 25742437[pmid], p. 52246. ISSN: 1940-087X.
- [10] Bernat Gel and Eduard Serra. “karyoploteR: an R/Bioconductor package to plot customizable genomes displaying arbitrary data”. In: *Bioinformatics* 33.19 (Oct. 2017), pp. 3088–3090. ISSN: 1367-4803. DOI: 10.1093/bioinformatics/btx346. URL: <https://doi.org/10.1093/bioinformatics/btx346> (visited on 08/12/2021).

- [11] The SAM/BAM Format Specification Working Group. *Sequence Alignment/Map Format Specification*. English. <https://samtools.github.io/hts-specs/SAMv1.pdf>. May 2018. (Visited on 08/09/2018).
- [12] Jennifer Harrow et al. “GENCODE: The reference human genome annotation for The ENCODE Project”. In: *Genome Res* 22.9 (Sept. 2012). 22955987[pmid], pp. 1760–1774. ISSN: 1088-9051.
- [13] Illumina. *Novaseq 6000*. <https://www.illumina.com/systems/sequencing-platforms/novaseq/specifications.html>.
- [14] Johnranganenome. *Biofinysics: How does bowtie2 assign MAPQ scores?* May 2014. URL: <http://biofinysics.blogspot.com/2014/05/how-does-bowtie2-assign-mapq-scores.html> (visited on 06/18/2021).
- [15] Mehran Karimzadeh et al. “Umap and Bismap: quantifying genome and methylome mappability”. In: *Nucleic Acids Research* (2018).
- [16] W. J. Kent et al. “BigWig and BigBed: enabling browsing of large distributed datasets”. In: *Bioinformatics* 26.17 (2010). <http://genome.ucsc.edu/>, pp. 2204–2207. eprint: /oup/backfile/content_public/journal/bioinformatics/26/17/10.1093_bioinformatics_btq351/2/btq351.pdf.
- [17] Felix Krueger. *FelixKrueger/Bismark*. original-date: 2015-11-07T18:14:13Z. July 2021. URL: <https://github.com/FelixKrueger/Bismark> (visited on 07/13/2021).
- [18] Felix Krueger and Simon R. Andrews. “Bismark: a flexible aligner and methylation caller for Bisulfite-Seq applications”. In: *Bioinformatics* 27.11 (2011), pp. 1571–1572. eprint: /oup/backfile/content_public/journal/bioinformatics/27/11/10.1093_bioinformatics_btr167/1/btr167.pdf.
- [19] Felix Krueger and Simon R. Andrews. “Bismark: a flexible aligner and methylation caller for Bisulfite-Seq applications”. en. In: *Bioinformatics* 27.11 (June 2011), pp. 1571–1572. ISSN: 1367-4803. DOI: 10.1093/bioinformatics/btr167. URL: <https://academic.oup.com/bioinformatics/article/27/11/1571/216956> (visited on 07/31/2019).
- [20] Melissa J Landrum et al. “ClinVar: improving access to variant interpretations and supporting evidence”. In: *Nucleic Acids Research* 46.D1 (2018), pp. D1062–D1067. eprint: /oup/backfile/content_public/journal/nar/46/d1/10.1093_nar_gkx1153/2/gkx1153.pdf.
- [21] Heng Li et al. “The Sequence Alignment/Map format and SAMtools”. In: *Bioinformatics* 25.16 (2009), pp. 2078–2079. eprint: /oup/backfile/content_public/journal/bioinformatics/25/16/10.1093_bioinformatics/btp352/2/btp352.pdf.
- [22] *marbl/CHM13*. original-date: 2019-02-28T16:00:16Z. July 2021. URL: <https://github.com/marbl/CHM13> (visited on 07/07/2021).
- [23] *Miniconda*. Apr. 2021. URL: <https://docs.conda.io/en/latest/>.
- [24] Shane Neph et al. “BEDOPS: high-performance genomic feature operations”. In: *Bioinformatics* 28.14 (2012), pp. 1919–1920. eprint: /oup/backfile/content_public/journal/bioinformatics/28/14/10.1093_bioinformatics_bts277/2/bts277.pdf.
- [25] Sergey Nurk et al. “The complete sequence of a human genome”. en. In: *bioRxiv* (May 2021). Publisher: Cold Spring Harbor Laboratory Section: New Results, p. 2021.05.26.445798. DOI: 10.1101/2021.05.26.445798. URL: <https://www.biorxiv.org/content/10.1101/2021.05.26.445798v1> (visited on 07/07/2021).
- [26] Nora Overlack et al. “Direct interaction of the Usher syndrome 1G protein SANS and myomegalin in the retina”. eng. In: *Biochimica Et Biophysica Acta* 1813.10 (Oct. 2011), pp. 1883–1892. ISSN: 0006-3002. DOI: 10.1016/j.bbamcr.2011.05.015.
- [27] B. S. Pedersen et al. “Fast and accurate alignment of long bisulfite-seq reads”. In: *ArXiv e-prints* (Jan. 2014). arXiv: 1401.1129 [q-bio.GN].

- [28] Judith A. Potashkin et al. “Bioinformatic Analysis Reveals Phosphodiesterase 4D-Interacting Protein as a Key Frontal Cortex Dementia Switch Gene”. eng. In: *International Journal of Molecular Sciences* 21.11 (May 2020), E3787. ISSN: 1422-0067. DOI: 10.3390/ijms21113787.
- [29] *QC Fail Sequencing : MAPQ values are really useful but their implementation is a mess.* en. URL: <https://sequencing.qcfail.com/articles/mapq-values-are-really-useful-but-their-implementation-is-a-mess> (visited on 06/18/2021).
- [30] Aaron R. Quinlan and Ira M. Hall. “BEDTools: a flexible suite of utilities for comparing genomic features”. In: *Bioinformatics* 26.6 (2010), pp. 841–842. eprint: /oup/backfile/content_public/journal/bioinformatics/26/6/10.1093/bioinformatics/btq033/3/btq033.pdf.
- [31] James T. Robinson et al. “Integrative genomics viewer”. en. In: *Nature Biotechnology* 29.1 (Jan. 2011), pp. 24–26. ISSN: 1546-1696. DOI: 10.1038/nbt.1754. URL: <https://www.nature.com/articles/nbt.1754> (visited on 08/09/2021).
- [32] Devon Ryan. *MethylDackel*. English. <https://github.com/dpryan79/MethylDackel>. (Visited on 08/09/2018).
- [33] Dirk Schübeler. “Function and information content of DNA methylation”. In: *Nature* 517 (Jan. 2015), 321 EP -.
- [34] Matija Snuderl et al. “Recurrent homozygous deletion of DROSHA and microduplication of PDE4DIP in pineoblastoma”. eng. In: *Nature Communications* 9.1 (July 2018), p. 2868. ISSN: 2041-1723. DOI: 10.1038/s41467-018-05029-3.
- [35] Artem Tarasov et al. “Sambamba: fast processing of NGS alignment formats”. In: *Bioinformatics* 31.12 (2015), pp. 2032–2034. DOI: 10.1093/bioinformatics/btv098. URL: <http://dx.doi.org/10.1093/bioinformatics/btv098>.
- [36] Romualdas Vaisvila et al. “Enzymatic methyl sequencing detects DNA methylation at single-base resolution from picograms of DNA”. en. In: *Genome Research* (June 2021). Company: Cold Spring Harbor Laboratory Press Distributor: Cold Spring Harbor Laboratory Press Institution: Cold Spring Harbor Laboratory Press Label: Cold Spring Harbor Laboratory Press Publisher: Cold Spring Harbor Lab. ISSN: 1088-9051, 1549-5469. DOI: 10.1101/gr.266551.120. URL: <https://genome.cshlp.org/content/early/2021/06/17/gr.266551.120> (visited on 07/13/2021).